MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, OPERATIONAL, ANALOG, KEY CODE DAA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic operational amplifier module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.116 failures per million operating hours for class I or 0.557 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DAA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/309-1 for class I and M28787/309-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 15.0 ±1.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 16.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

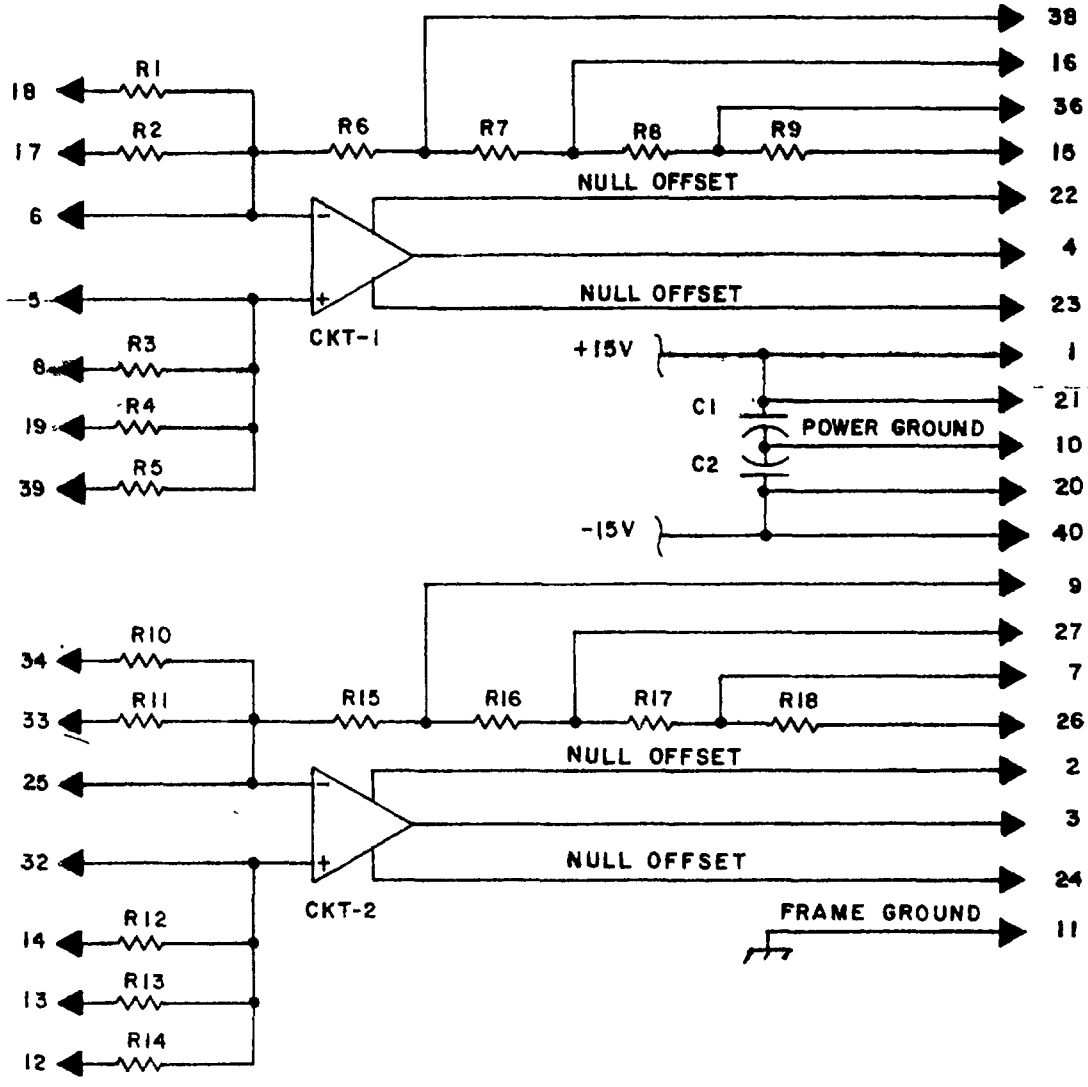
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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NOTES
1. R1 = R2 = R3 = R4 = R6 = R10 = R11 = R12 = R13 = R15 = 10 kΩ ±1%, 1/20 W.
2. R5 = R8 = R14 = R17 = 40.2 kΩ ±1%, 1/20 W.
3. R7 = R16 = 20 kΩ ±1%, 1/20 W.
4. R9 = R18 = 80.6 kΩ ±1%, 1/20 W.
5. C1 = C2 = 0.1 µF ±20%, BX characteristic ceramic.
6. Pins 28, 29, 30, 31, 35 and 37 are unused.

FIGURE 1. DAA module function diagram, operational amplifier.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	V_S stress	II	Q	-90	90									mV
B1-B4	Output short circuit current	III	S	9	40	9	55	9	40	9	55	9	40	mA
C1-C7	Isolation													
	Initial	IV	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
D1,D2	Power supply current	V	S	0.5	6.2	0.5	6.5	0.5	6.0	0.5	7.0	0.5	5.5	mA
E1,E2	Power supply filter capacitance	VI	X	0.08	0.12	0.08	0.12	0.08	0.12	0.08	0.12	0.08	0.12	µF
F1-F10	Resistance													
	Initial	VII	X	9.90	10.10	9.85	10.15	9.85	10.15	9.82	10.18	9.84	10.16	kΩ
	End-of-life	VII	X	9.85	10.15	9.80	10.20	9.80	10.21	9.77	10.23	9.77	10.23	kΩ
F11-F14	Resistance													
	Initial	VII	X	39.80	40.60	39.60	40.80	39.60	40.80	39.48	40.92	39.56	40.84	kΩ
	End-of-life	VII	X	39.60	40.80	39.40	41.00	39.40	41.00	39.27	41.12	39.27	41.12	kΩ
F15,16	Resistance													
	Initial	VII	X	19.80	20.20	19.70	20.30	19.70	20.30	19.64	20.36	19.68	20.32	kΩ
	End-of-life	VII	X	19.70	20.30	19.60	20.40	19.60	20.40	19.54	20.46	19.54	20.46	kΩ
F17,F18	Resistance													
	Initial	VII	X	79.79	81.41	79.39	81.81	79.39	81.81	79.15	82.05	79.31	81.89	kΩ
	End-of-life	VII	X	79.39	81.81	78.99	82.21	78.99	32.21	78.75	82.45	78.75	82.45	kΩ
G1,G2	Input offset voltage check	VIII	X	-3.0	3.0	-3.5	3.5	-3.5	3.5	-4.0	4.0	-4.0	4.0	mV
H1,H2	Input offset current check	VIII	X	-30	30	-43	43	-37	37	-70	70	-45	45	nA
J1,J2	Input bias current check	IX	X	1.0	110	1.0	265	1.0	110	1.0	265	1.0	110	nA
K1,K2	AC gain (1) functional check													
	Initial	X	X	0.980	1.020	0.970	1.030	0.970	1.030	0.962	1.038	0.962	1.038	V/V
	End-of-life	X	X	0.970	1.080	0.960	1.040	0.960	1.040	0.952	1.048	0.952	1.048	V/V

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K3,K4	AC gain (2) functional check													
	Initial	X	X	1.960	2.040	1.940	2.060	1.940	2.060	1.924	2.076	1.924	2.076	V/V
	End-of-life	X	X	1.940	2.060	1.920	2.080	1.920	2.080	1.904	2.096	1.904	2.096	V/V
K5,K6	AC gain (4.02) functional check													
	Initial	X	X	3.940	4.100	3.899	4.141	3.899	4.141	3.867	4.173	3.867	4.173	V/V
	End-of-life	X	X	3.899	4.141	3.859	4.181	3.859	4.181	3.827	4.213	3.827	4.213	V/V
K7,K8	AC gain (8.06) funcitonal check													
	Initial	X	X	7.899	8.221	7.818	8.302	7.818	8.302	7.754	8.366	7.754	8.366	V/V
	End-of-life	X	X	7.818	8.302	7.738	8.382	7.738	8.832	7.673	8.447	7.673	8.447	V/V
L1,L2	Small signal bandwidth	X	S	0.438	5.000	0.460	5.000	0.403	5.000	0.493	5.000	0.372	5.000	MHz
M1,M2	Gain peaking ratio check	X	S	1.00	1.35	1.00	1.35	1.00	1.35	1.00	1.35	1.00	1.35	V/V
N1,N2	Large signal voltage gain check	XI	X	50	700	25	700	25	700	25	700	25	700	V/mV
P1-P4	Slew rate check	XII	X	0.30	10.0	0.29	10.0	0.30	10.0	0.24	10.0	0.30	10.0	V/µS
R1,R2	Maximum output voltage	XIII	S	6.00	9.55	6.00	9.55	6.00	9.55	6.00	9.55	6.00	9.55	V rms
S1,S2	Common mode rejection ratio check	XIV	X	∞	-40	∞	-40	∞	-40	∞	-40	∞	-40	dB
T1,T2	Power supply rejection ratio check	XV	S	∞	-52	∞	-52	∞	-52	∞	-52	∞	-52	dB
U1-U3	Continuity	XVI	S	0	1.0	0	1.0	0	1.0	0	1.0	0	1.0	Ω
V1,V2	Output latch check (end-of-life)	Fig 4	X	---	---	---	---	-172	172	---	---	-187	187	mV

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B and D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4. Following the life test, and prior to disconnecting the power, electrical tests V1 through V2 shall be performed in accordance with figure 4. While still at the maximum class temperature, electrical tests D through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through U shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Cross reference of classification. Military Specification MIL-M-28787/309 supersedes Naval Electronic Systems Command Drawing No. 28687-0102-887. Part number M28787/309-1 supersedes part number 0102-887.

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* 6.5 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (±16.5 V dc supply, no signal)	107.25	115.5	mW
Input offset voltage	±3.5 max	±4.0 max	mV
Input offset current	±43 max	±70	nA
Input bias current (average, each input)	±265	±265	nA
Common mode rejection ratio	-40 max	-40 max	dB
Power supply rejection ratio	-52 max	-52 max	dB
Slew rate (unity gain) ($R_L \geq 2$ kΩ, V_{CC} = ±15 V)	0.29 min	0.24 min	V/µs
Small signal bandwidth (unity gain)	0.403 min	0.372 min	MHz
Maximum output voltage (≤ 1% distortion) (R_L = 2 kΩ, f = 1.0 kHz, V_{CC} = 13.5 V dc)	6.00 min	6.00 min	V rms
Output drive current capability (V_O = 8.5 V_p)	4.25 min	4.25 min	mA
Large signal voltage gain	25,000 min	25,000 min	V/V
Gain options	-9.17 dB min to 29.59 max		
Gain accuracy (R_L = 2 kΩ)	±4.0	±4.8	

* 6.6 Module function. This module contains two independent general purpose amplifiers and two 0.1 µF decoupling capacitors as illustrated on figure 1.

6.6.1 Parallel outputs. Paralleling of outputs is not permissible.

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

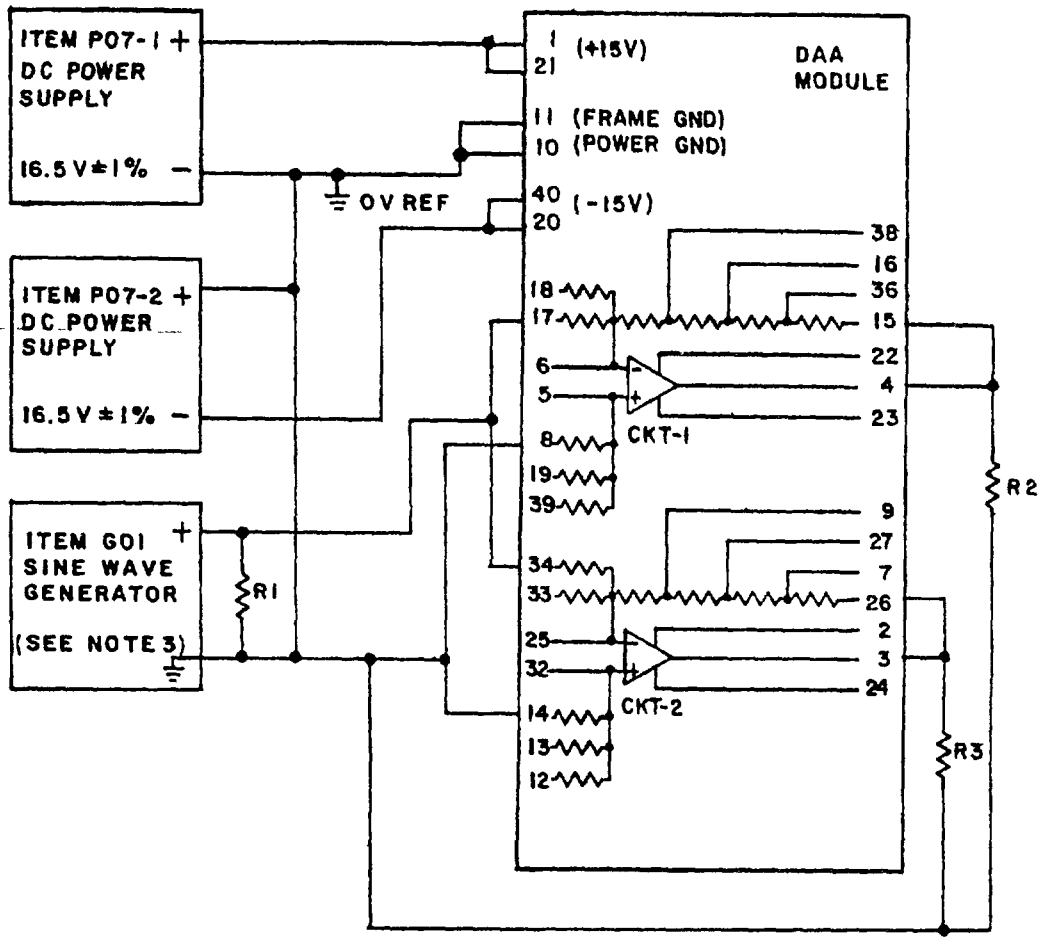
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NOTES:
1. R1 = 49.9Ω ±5%, 1 W.
2. R2 = R3 = 2 kΩ ±1%, 1/8 W.
3. Item G01 setup:
   Amplitude = 2 V_{pp} ±5%.
   Frequency = 1 kHz ±2%.

FIGURE 2. Operating temperature and component temperature test circuit.

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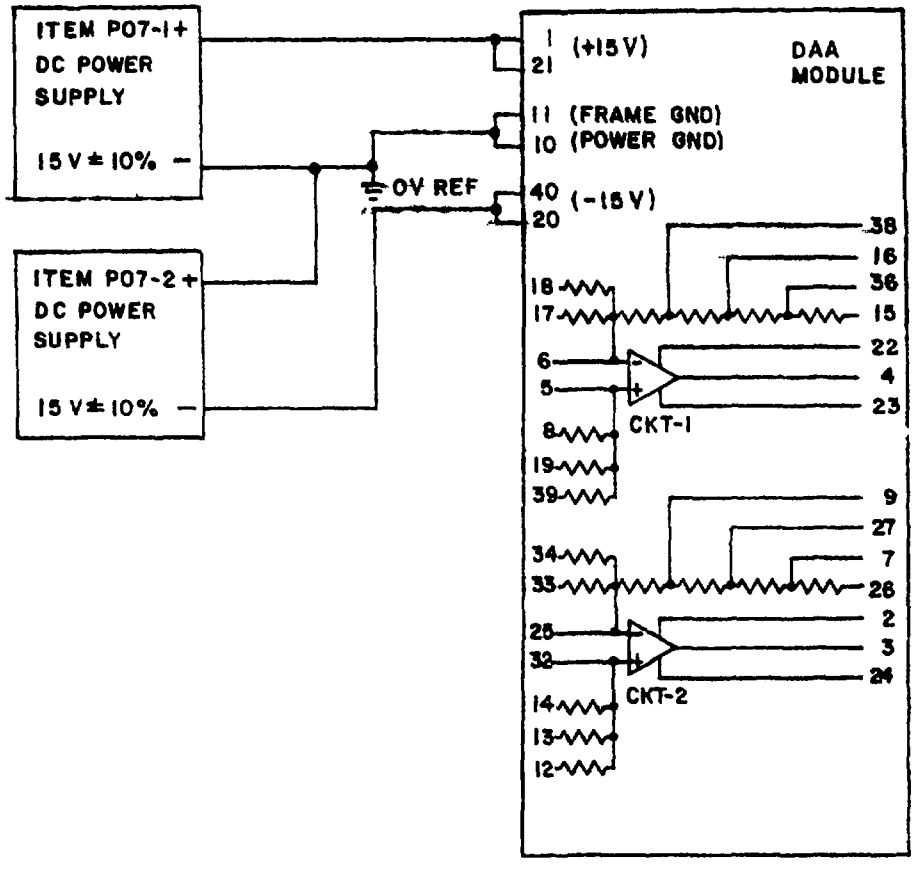
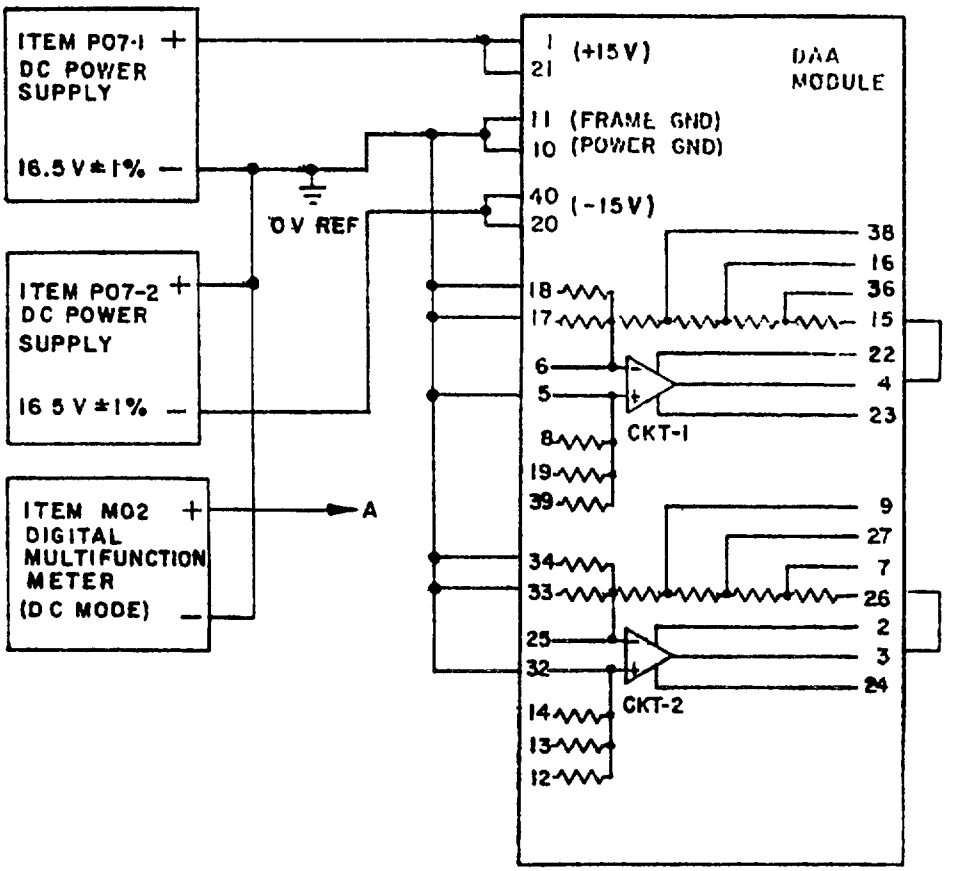


FIGURE 3. Vibration test circuit.

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NOTES.
1. Following the life test period, prior to disconnecting the power, connect point A to pin 4. The voltage shall meet the limits specified in table I for test condition V1.
2. For test condition V2, connect point A to pin 3.
3. Test condition V verifies that the amplifiers do not latchup.

FIGURE 4 Life test circuit.

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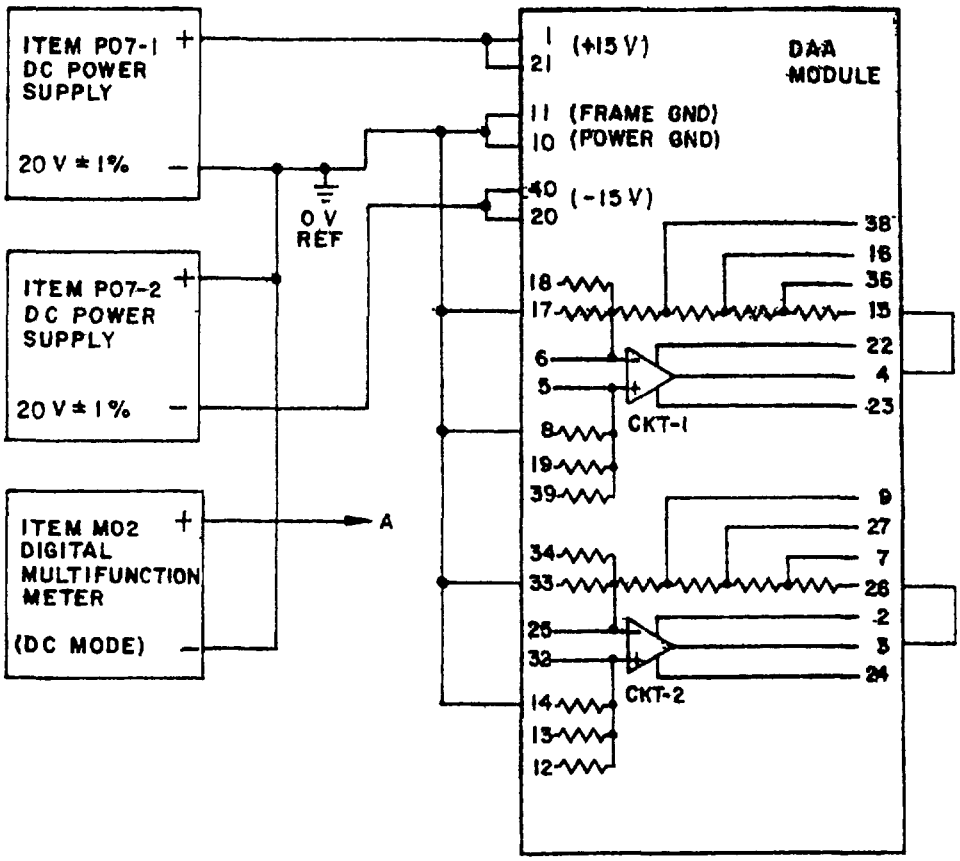
TABLE II. V_S stress test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	V_S stress	4	-90	90	mV	5
A2	V_S stress	3	-90	90	mV	5

TABLE III. Output short circuit current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Output short circuit current	Open	4,15	9	40	mA	6
B2	↓	4,15	Open	↓	↓	↓	↓
B3	↓	Open	3,26	↓	↓	↓	↓
B4	↓	3,26	Open	↓	↓	↓	↓

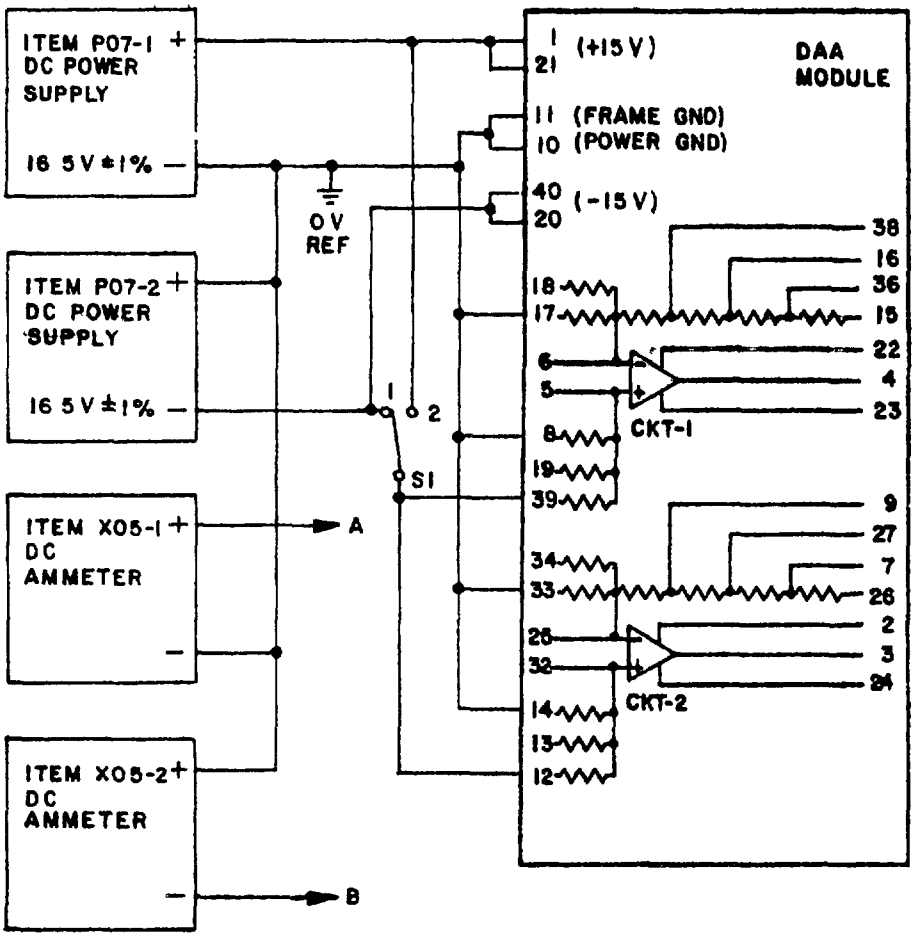
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NOTES
1. Measurement shall be recorded after power has been applied for a minimum of 1 minute.
2. This test shall be performed at 25° ±5°C only.

* FIGURE 5. V_S stress test circuit for table II.

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NOTES:
1. S1 = SPDT switch.
2. Place switch S1 in position 1 for test conditions B1 and B3 and in position 2 for test conditions B2 and B4.
3. With points A and B disconnected, item X05 should indicate zero current on the 200 µA range.
4. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken after 1 minute.
5. Since items X05-1 and X05-2 are not used simultaneously, only one item X05 is required.

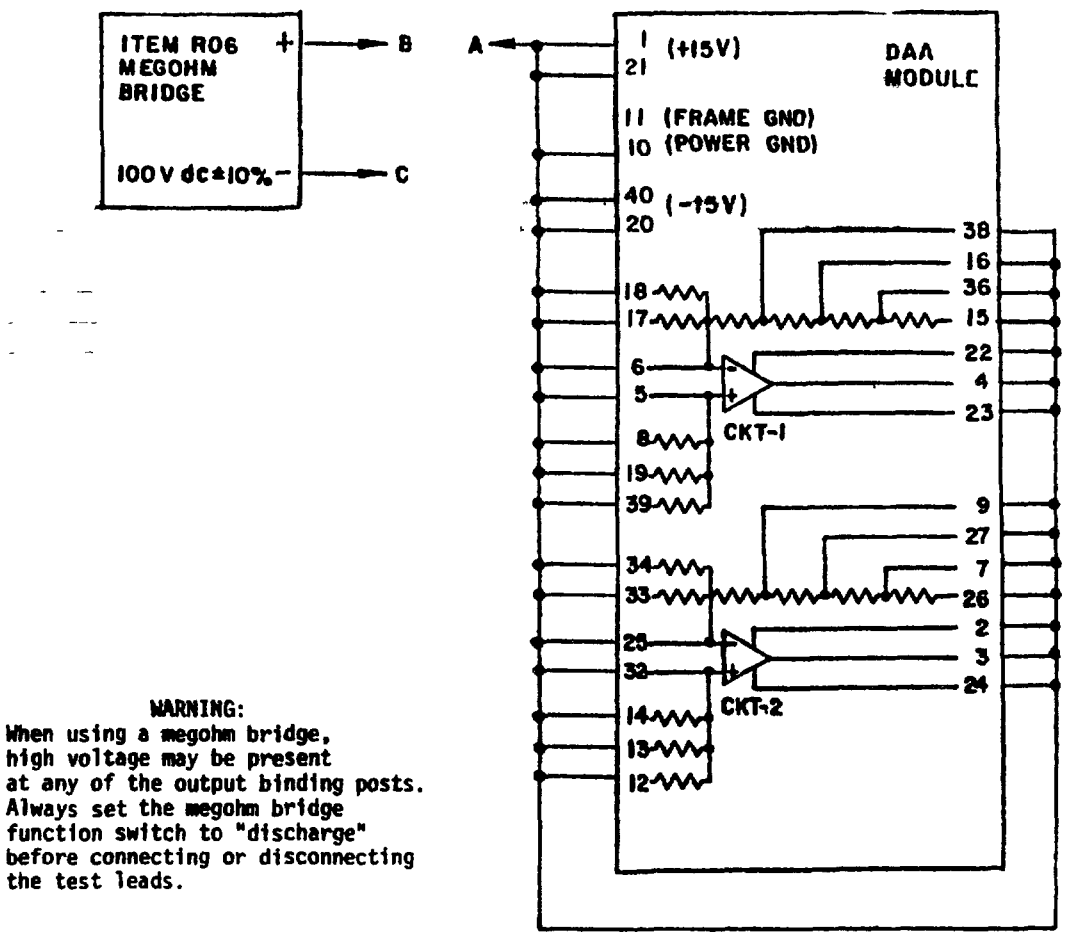
* FIGURE 6. Output short circuit current test circuit for table III.

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* TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		B to pin	C to pin	Min	Max		
C1	Isolation, module pins to frame	28,29,30,A, 31,35,37	11	10	∞	MΩ	7
C2	Isolation, used to unused pins	A,11	28,29,30, 31,35,37	↓	↓	↓	↓
C3	Isolation, unused to unused pins	28	29,30,31, 35,37				
C4	↓	29	30,31,35, 37				
C5		30	31,35,37				
C6		31	35,37				
C7		35	37				

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WARNING:
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 7. Isolation test circuit for table IV.

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TABLE V. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
D1	Power supply current	0.5	6.2	mA	8
D2	Power supply current	0.5	6.2	mA	8

TABLE VI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	C1 Power supply filter capacitance	1	10	0.08	0.12	µF	9
E2	C2 Power supply filter capacitance	20	10	0.08	0.12	µF	9

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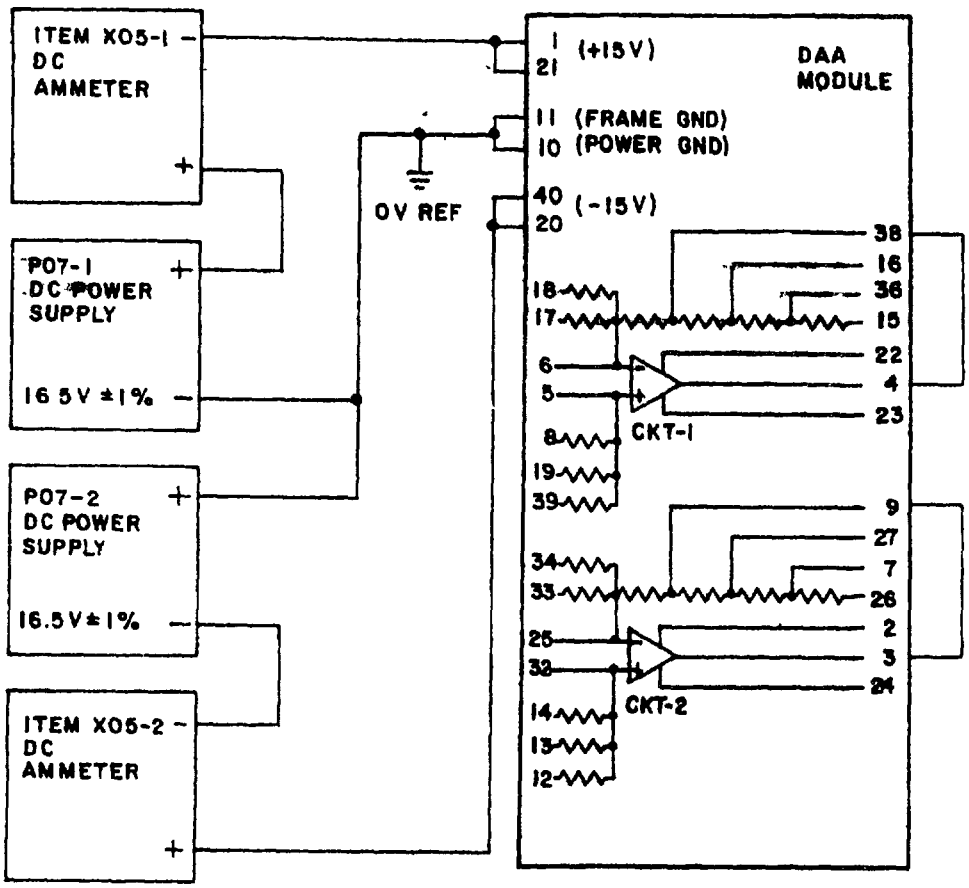
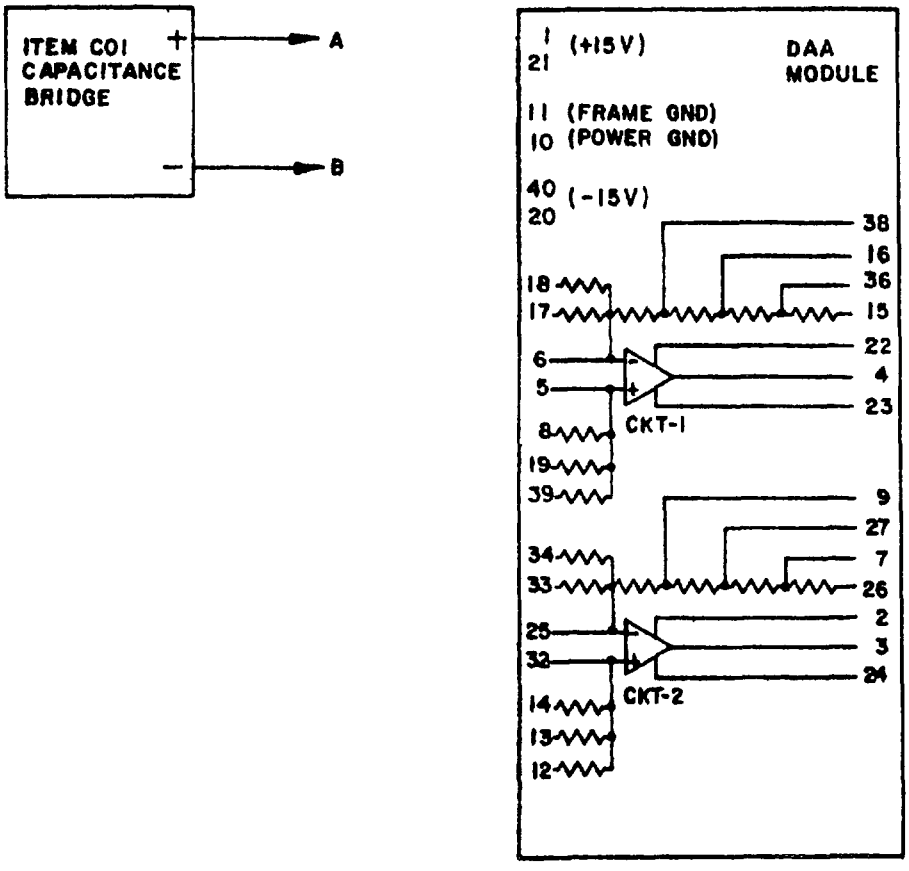


FIGURE 8. Power supply current test circuit for table V

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NOTE: Item C01 setup:
Frequency = 1 kHz ±10%.
Amplitude = 0.3 V_{pk} ±10%.

FIGURE 9. Power supply filter capacitance test circuit for table VI.

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TABLE VII. Resistance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	R1 Resistance	18	6	9.90	10.10	kΩ	10
F2	R2	17	6				
F3	R6	38	6				
F4	R3	8	5				
F5	R4	19	5				
F6	R10	34	25				
F7	R11	33	25				
F8	R15	9	25				
F9	R12	14	32				
F10	R13	13	32				
F11	R14	12	32	39.80	40.60		
F12	R17	27	7				
F13	R5	39	5				
F14	R8	16	36				
F15	R7	16	38	19.80	20.20		
F16	R16	27	9	19.80	20.20		
F17	R18	26	7	79.79	81.41		
F18	R9	15	36	79.79	81.41		

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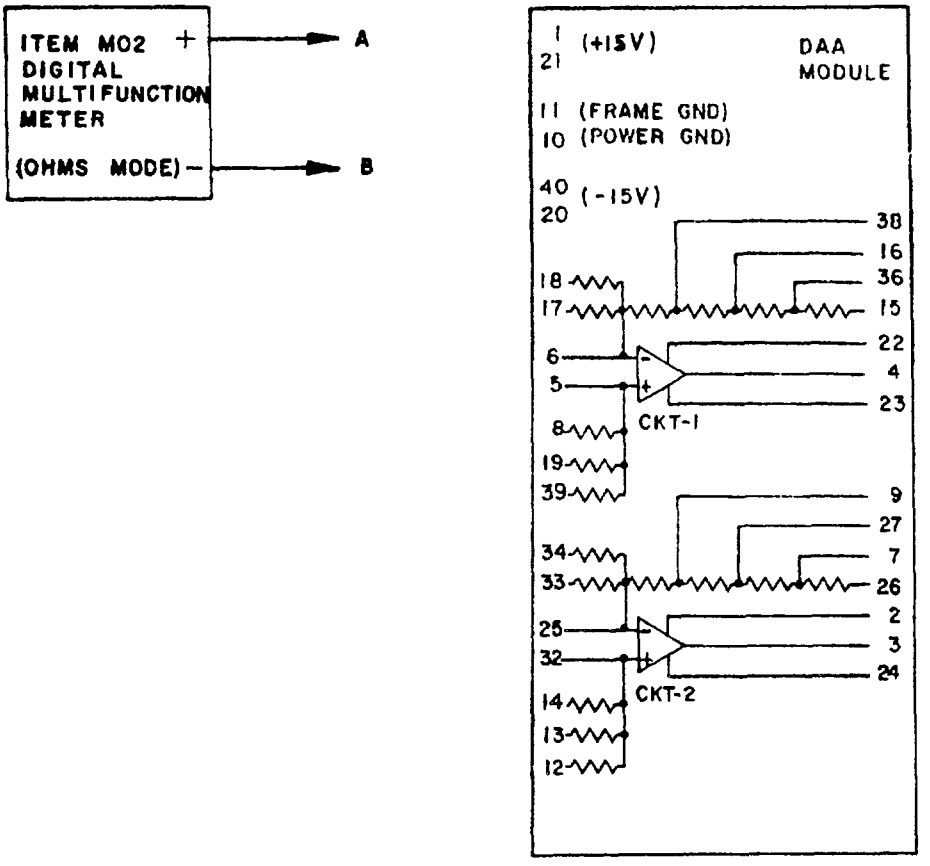


FIGURE 10. Resistance test circuit for table VII.

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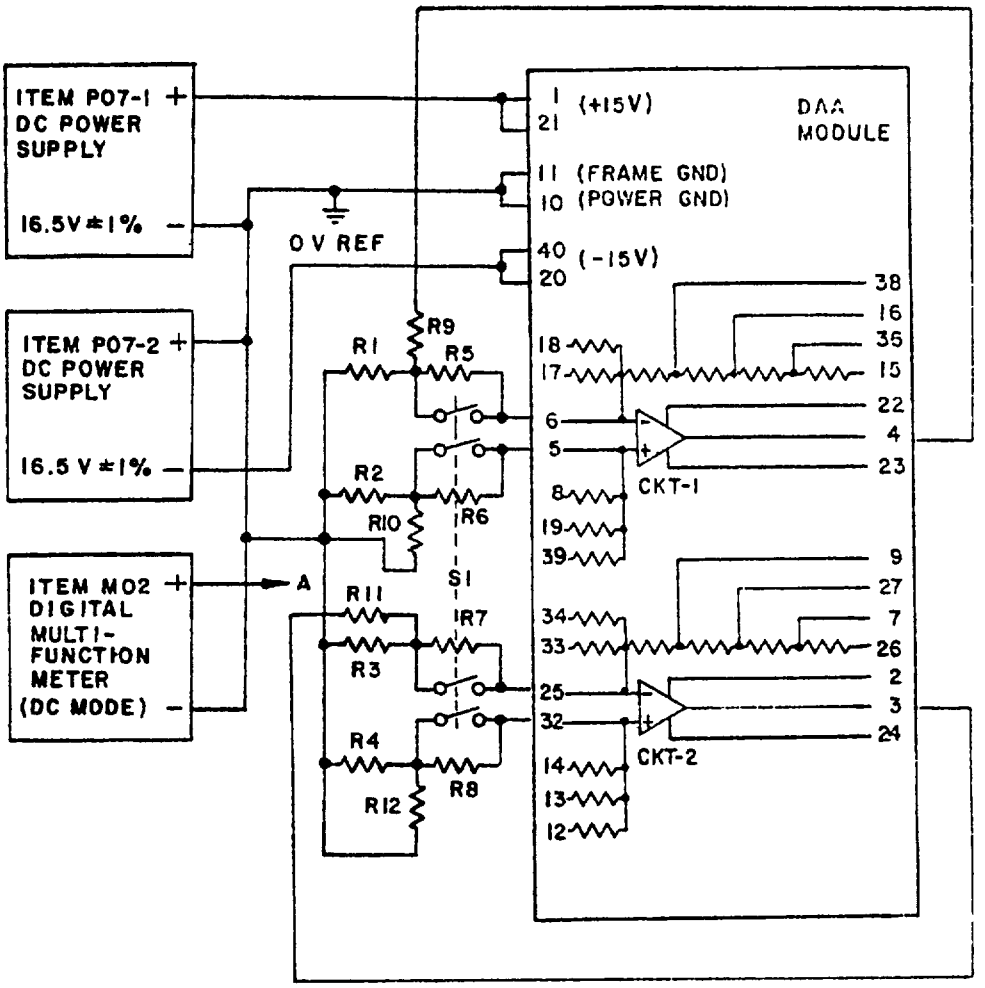
TABLE VIII. Input offset voltage and input offset current check test. 1/ 2/

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
G1	Input offset voltage check	4	-300	300	mV	11
G2	Input offset voltage check	3	-300	300		
H1	Input offset current check (test condition G1)-(Item M02 reading)	4	-30	30		
H2	Input offset current check (test condition G2)-(Item M02 reading)	3	-30	30		

1/ $V_{IO} = \frac{V_{OUT}}{\frac{R_g + R_1}{R_1}} \approx \frac{V_{OUT}}{100}$

2/ $I_{IO} = \frac{\Delta V_{OUT}}{\text{Gain} \times R_{Series}} = \frac{\text{(Test condition G)-(Item M02 reading)}}{100 \times 10\ k\Omega}$

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NOTES:
1. R1 = R2 = R3 = R4 = 50Ω ±0.1%, 1/8 W.
2. R5 = R6 = R7 = R8 = 10 kΩ ±0.1%, 1/8 W.
3. R9 = R10 = R11 = R12 = 5 kΩ ±1%, 1/8 W.
4. S1 = 4 PST switch.
5. Switch S1 is closed for test conditions G1 and G2 and is open for test conditions H1 and H2.

FIGURE 11. Input offset voltage and input offset current check test circuit for table VIII.

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TABLE IX. Input bias current check test. 1/ 2/

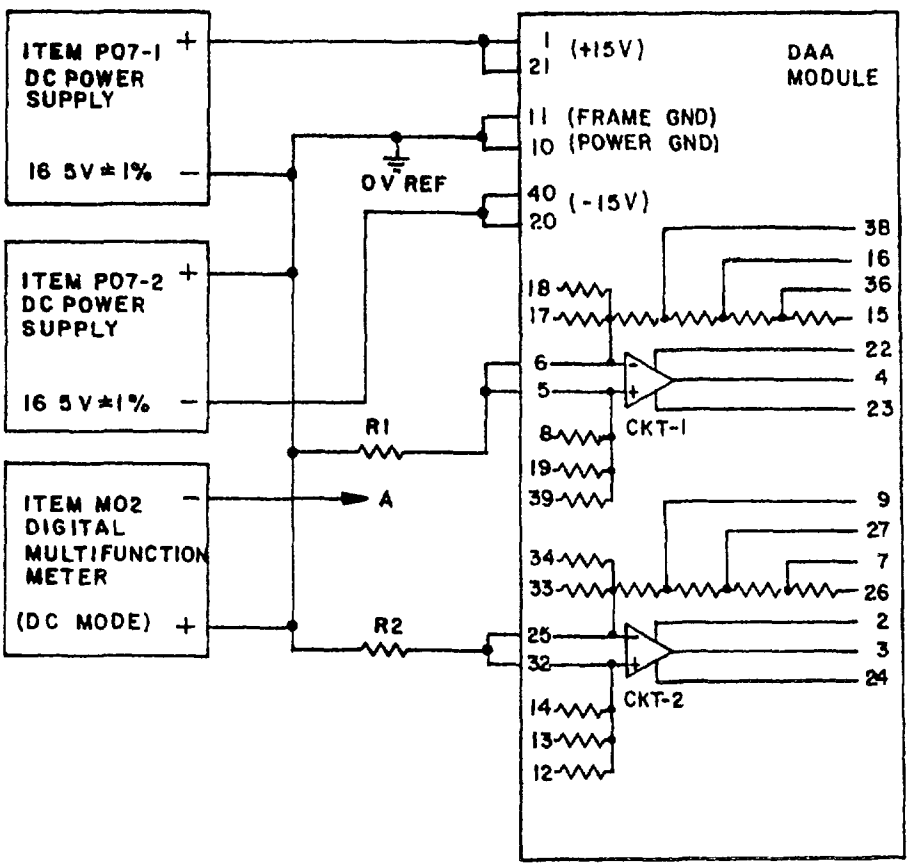
Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J1	Input bias current check	5	0.2	22.0	mV	12
J2	Input bias current check	25	0.2	22.0	mV	12

1/ Average input bias current (table I) = $\frac{\text{Test condition J}}{(100\ \text{k}\Omega)\ X2}$.

2/ This test in conjunction with test condition H (offset current test) verifies an input bias current ≤ 125 nA per input at 25°C initial where:

$$I_{IB}\ (max) = I_{IB}(avg) + \frac{|I_{IO}|}{2}.$$

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NOTE R1 = R2 = 100 kΩ ±0.1%, 1/8 W.

FIGURE 12. Input bias current check test circuit for table IX.

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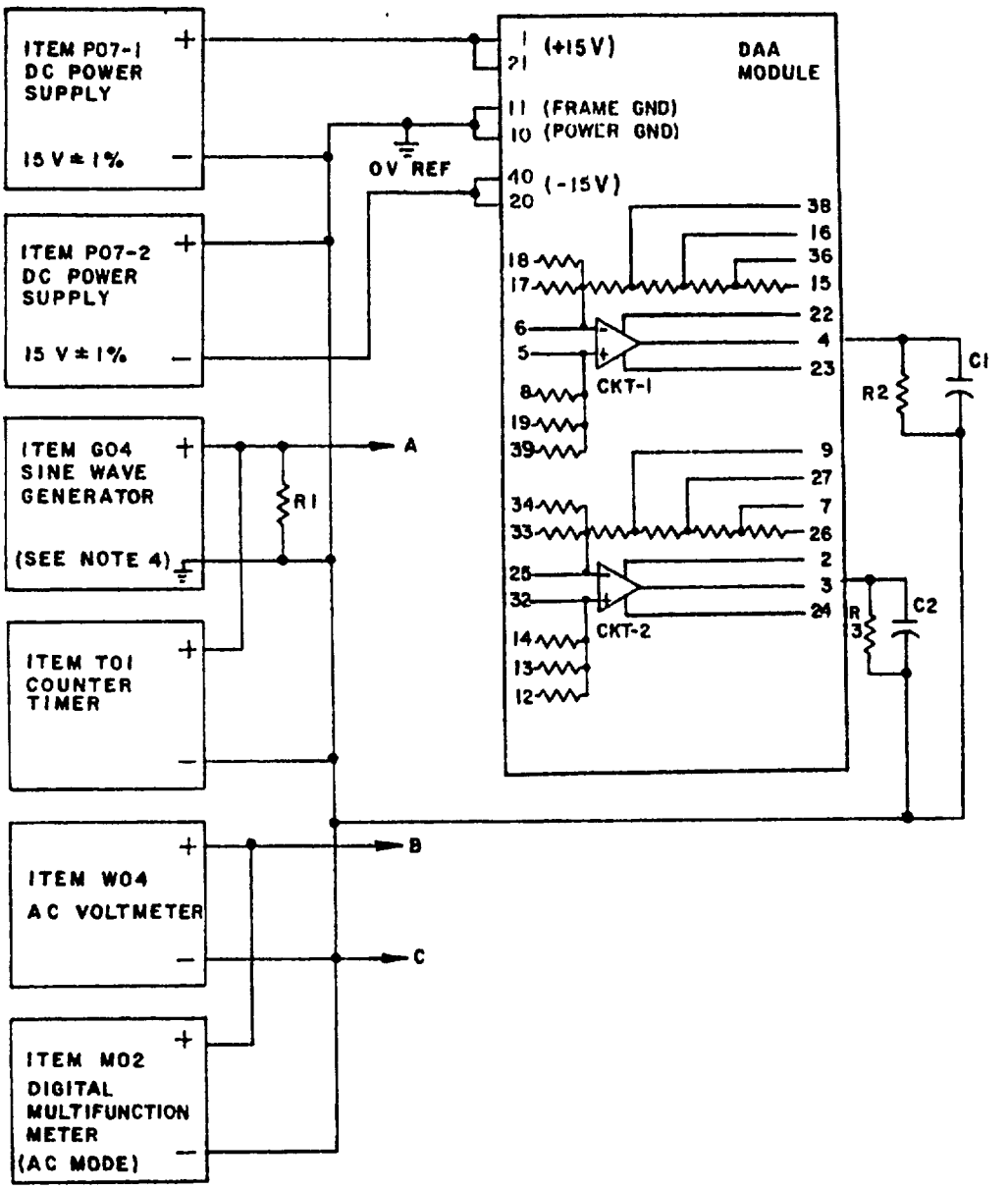
TABLE X. AC gain functional check, small signal bandwidth and gain peaking ratio check test. 1/ 2/

Test condition	Parameter	Connect			Short pins	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin		Min	Max		
K1	AC gain (1) functional check	17	4,38	8,19		0.980	1.020	V rms	13
K2	AC gain (1) functional check	33	3,9	14,13		0.980	1.020		
K3	AC gain (2) functional check	17	4,16	8,19	38,6	1.960	2.040		
K4	AC gain (2) functional check	33	3,27	14 13	9,25	1.960	2.040		
K5	AC gain (4.02) functional check	17	4,36	8,39	16,6	3.940	4.100		
K6	AC gain (4.02) functional check	33	3,7	14,12	27,25	3.940	4.100		
K7	AC gain (8.06) functional check	17	4,15	8,39	36,6	7.899	8.221		
K8	AC gain (8.06) functional check	33	3,26	14,12	7,25	7.899	8.221		
L1	Small signal bandwidth	5	4,6	Open		0.438	5.000	MHz	
L2	Small signal bandwidth	32	3,26			0.438	5.000	MHz	
M1	Gain peaking ratio check	5	4,6			20	27	mV rms	
M2	Gain peaking ratio check	32	3,25			20	27	mV rms	

1/ AC gain = $\frac{V_{OUT}}{V_{IN}} = \frac{V_{OUT}}{1\ V\ rms}$ = Item M02 reading.

2/ Gain peaking ratio = $\frac{V_{OUT}}{V_{IN}} = \frac{V_{OUT}}{20\ mV\ rms} = \frac{\text{Item W04 reading}}{20\ mV\ rms}$.

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NOTES:
1. R1 = 49.9Ω ±5%, 1 W.
2. R2 = R3 = 2 kΩ ±1%, 1/8 W.
3. C1 = C2 = 100 pF ±20%.
4. For test condition K, adjust item G04 1.0 Vrms ±0.1% at 1.0 kHz ±1% at point A with A connected and read item M02.
5. Using item W04, adjust item G04 to 20 mVrms ±1% at 1.0 kHz ±1% at point B as connected per test condition L1.
6. For test conditions L1 and L2, maintain input voltage within ±1% while increasing the frequency until output rolls off to 14.14 mVrms (-3 dB) as measured on item W04. Item G04 frequency at this -3 dB point shall conform to table X limits as measured on item T01.
7. For test conditions M1 and M2, maintain input voltage at 20 mVrms ±1% while increasing the frequency to 5 MHz. Maximum output voltage on item W04 shall conform to table X limits.
8. The gain peaking ration of 1.35 verifies a minimum phase margin $\Phi m = 45^{\circ}$.

FIGURE 13. AC gain functional check, small signal bandwidth and gain peaking ratio check test circuit for table X.

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TABLE XI. Large signal voltage gain check test. 1/

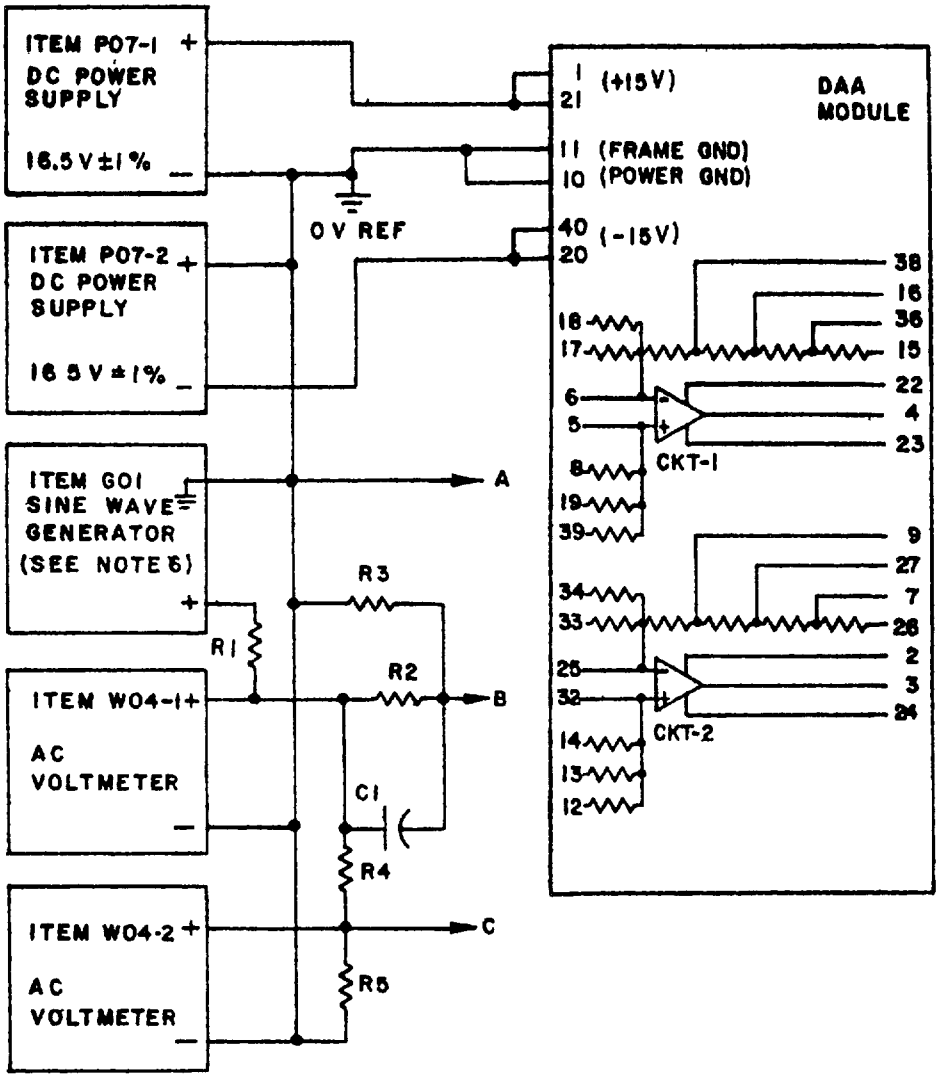
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Large signal voltage gain check	5	6	4	1.0	14.0	mV rms	14
N2	Large signal voltage gain check	32	25	3	1.0	14.0	mV rms	14

1/ $\text{Gain} = \dfrac{-V_{OUT}\left(\frac{R_3 + R_2}{R3}\right)}{V_{(R3+R2)}} \approx \dfrac{-(7.0\ \text{Vrms})\ (100)}{\text{Test condition N}}$.

TABLE XII. Slew rate check test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Slew rate check (neg. to pos.)	17	4	0.4	13.3	µs	15,16
P2	Slew rate check (pos. to neg.)	17	4	↓	↓	↓	↓
P3	Slew rate check (neg. to pos.)	33	3				
P4	Slew rate check (pos. to neg.)	33	3				

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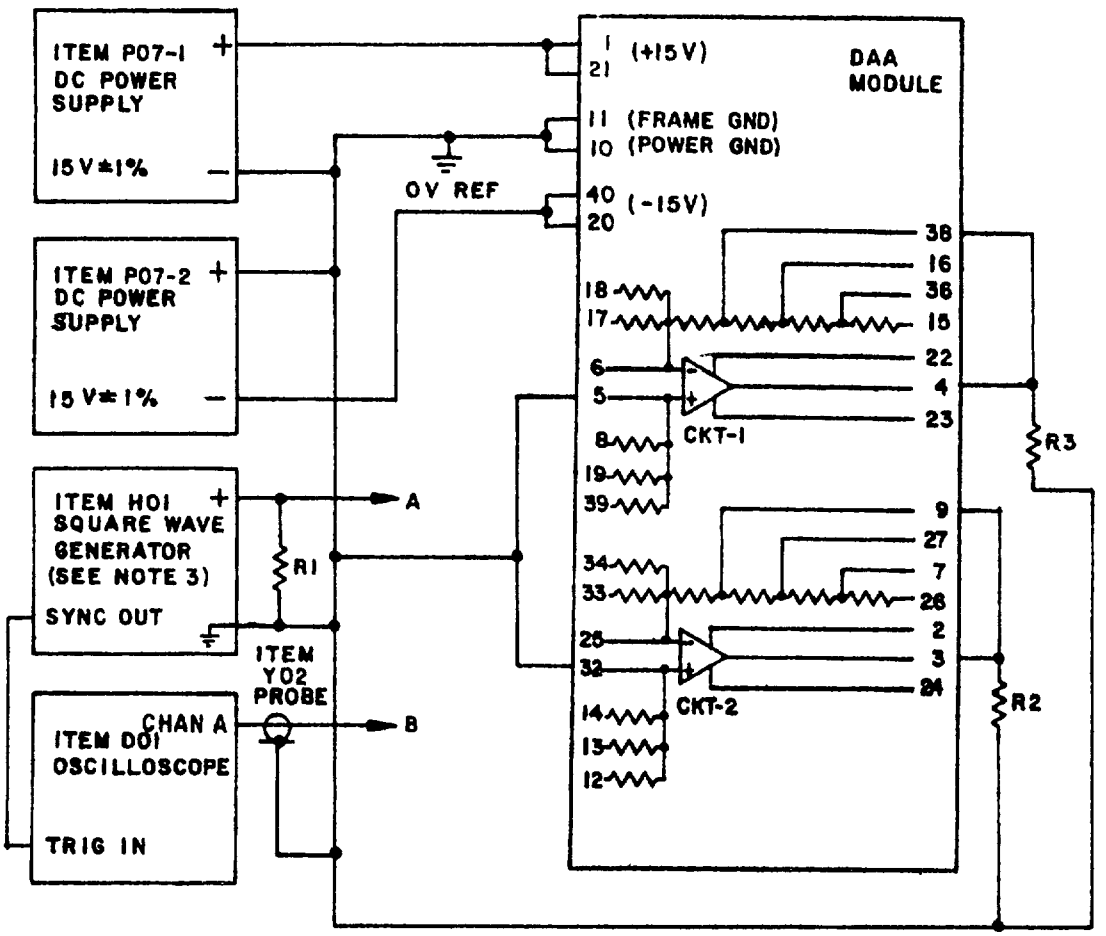
NOTES:
1. R1 = R4 = 49.9 kΩ ±1%, 1/8 W.
2. R2 = 100 kΩ ±1%, 1/8 W.
3. R3 = 1.0 kΩ ±1%, 1/8 W.
4. R5 = 10 kΩ ±1%, 1/4 W.
5. C1 = 470 pF ±10%, 50 Vdc.
6. For each test condition:

   a. Adjust item G01 frequency to 10 Hz ±5% and amplitude to 7.0 Vrms ±1% at point C as measured on item W04-2 with points A, B, C connected.
   b. Read item W04-1.

7. Since items W04-1 and W04-2 are not used simultaneously only one item W04 is required.
8. Test condition N verifies a minimum large signal voltage gain at 25°C initial of 50,000.

FIGURE 14. Large signal voltage gain check test circuit for table XI.

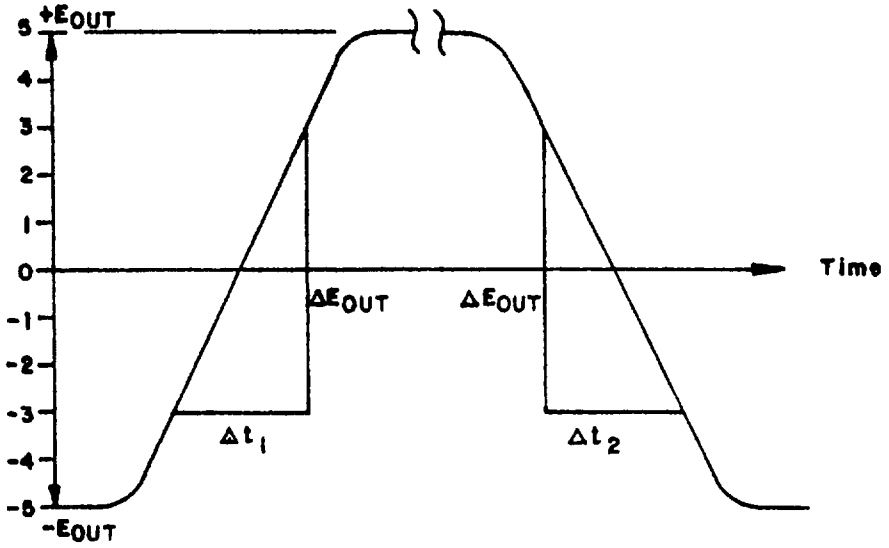
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NOTES:
1. R1 = 49.9Ω ±5%, 1 W.
2. R2 = R3 = 2 kΩ ±1%, 1/8 W.
3. Item H01 setup:
   Frequency = 1 kHz ±1%.
   Amplitude = 10 V_{pp} ±10% at point A.
4. This test verifies a slew rate of 0.3 V/µs minimum at 25°C initial.

FIGURE 15. Slew rate check test circuit for table XII.

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NOTES:

1. Slew rate = $\frac{\Delta E_{OUT}}{\Delta t} = \frac{4\ V}{\Delta t}$.

   Δt = 4 V/0.3 V/µs = 13.3 µs maximum.

2. Measure Δt_1 for P1 and P3 and measure Δt_2 for P2 and P4.

FIGURE 16. Slew rate check test circuit waveform for table XII.

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TABLE XIII. Maximum output voltage test.

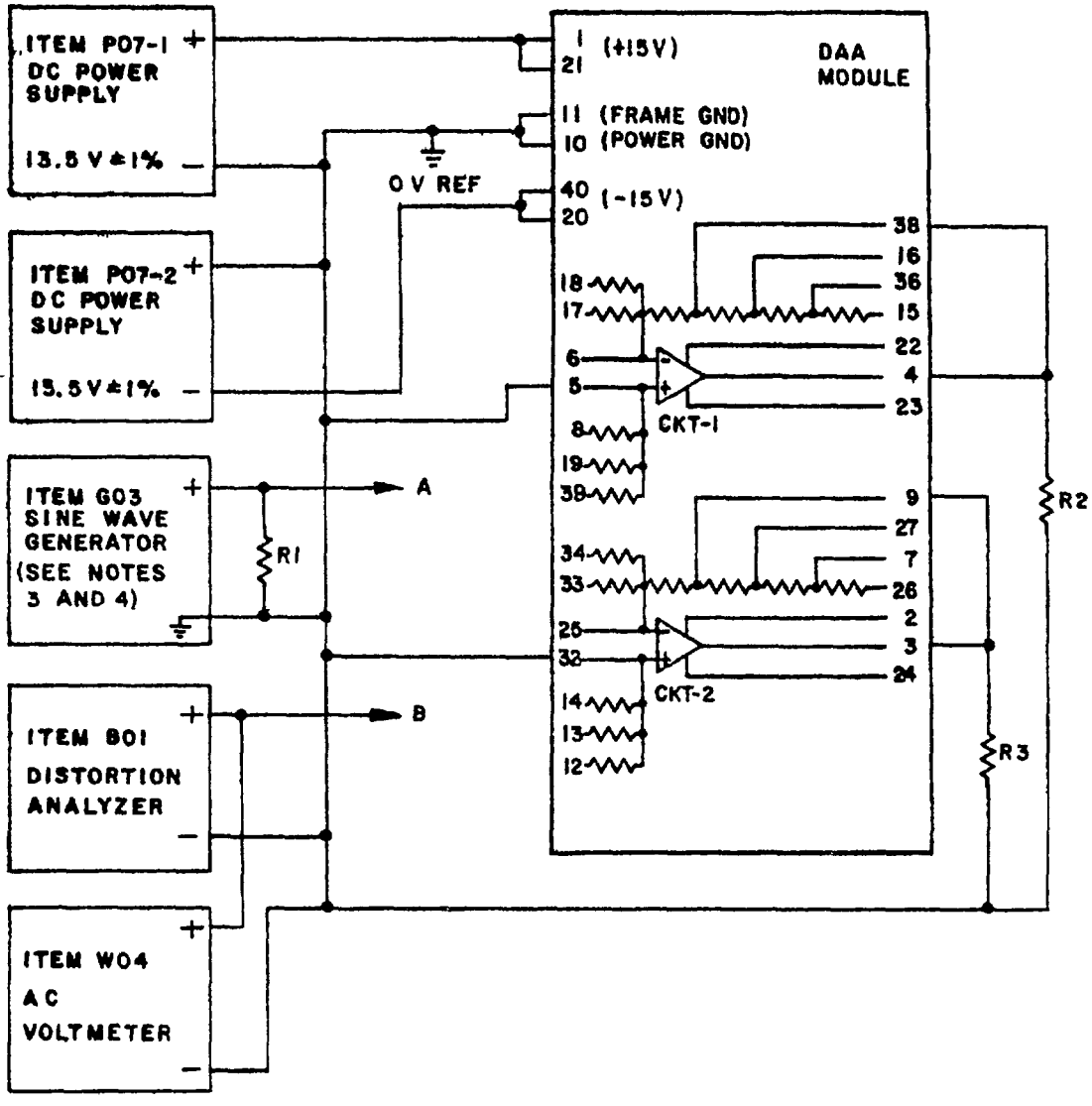
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Maximum output voltage	17	4	6.00	9.55	V rms	17
R2	Maximum output voltage	33	3	6.00	9.55	V rms	17

TABLE XIV. Common mode rejection ratio check test. 1/

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S1	Common mode rejection ratio check	17,8	4	0	10	mV rms	18
S2	Common mode rejection ratio check	33,14	3	0	10	mV rms	18

1/ $CMRR = 20 \log \left(\frac{V_{OUT.}}{V_{IN}}\right) = 20 \log \left(\frac{\text{Item W05 reading}}{1.0 \text{ V rms}}\right)$.

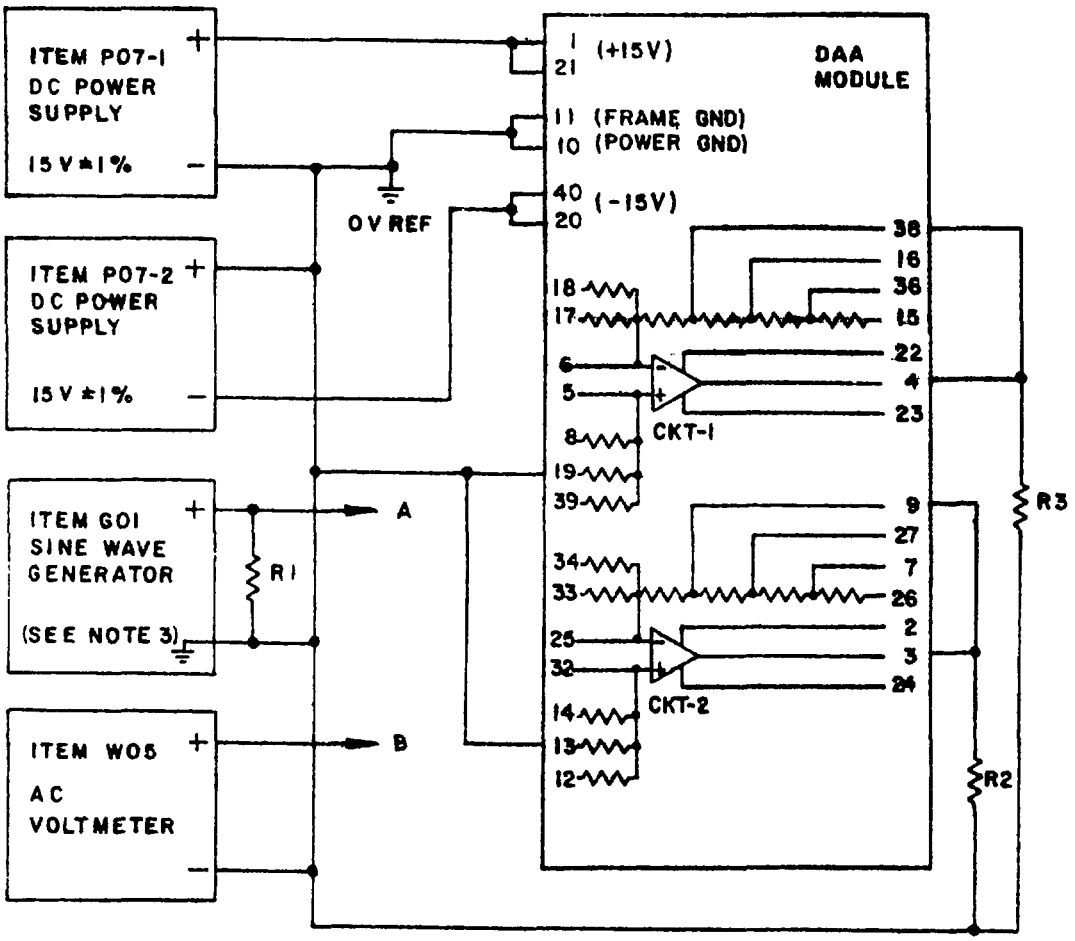
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NOTES:
1. R1 = 200Ω ±5%, 1 W.
2. R2 = R3 = 2 kΩ ±1%, 1/8 W.
3. Item G08 frequency = 1.0 kHz ±1% and amplitude = 0.5 Vrms ±10%.
4. For test condition R1, increase item G08 amplitude until the distortion (on item B01) equals 1.0 ±0.1%. The reading on item W05 shall meet the limits specified in table XIII.
5. Repeat note 4 for test condition R2.
6. Test condition R verifies a minimum peak output voltage of 8.5 V, a minimum output drive capability of 4.25 mA, and a maximum module distortion of 1%.

* FIGURE 17 Maximum output voltage test circuit for table XIII.

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NOTES.
1. R1 = 49.9Ω ±5%, 1 W.
2. R2 = R3 = 2 kΩ ±1%, 1/8 W.
3. Item G01 setup:
   Amplitude at A = 1.0 Vrms ±1% with A connected per table XIV.
   Frequency = 1 kHz ±1%.
4. This test verifies a common mode rejection ratio of -40 dB maximum.

FIGURE 18 Common mode rejection ratio check test circuit for table XIV.

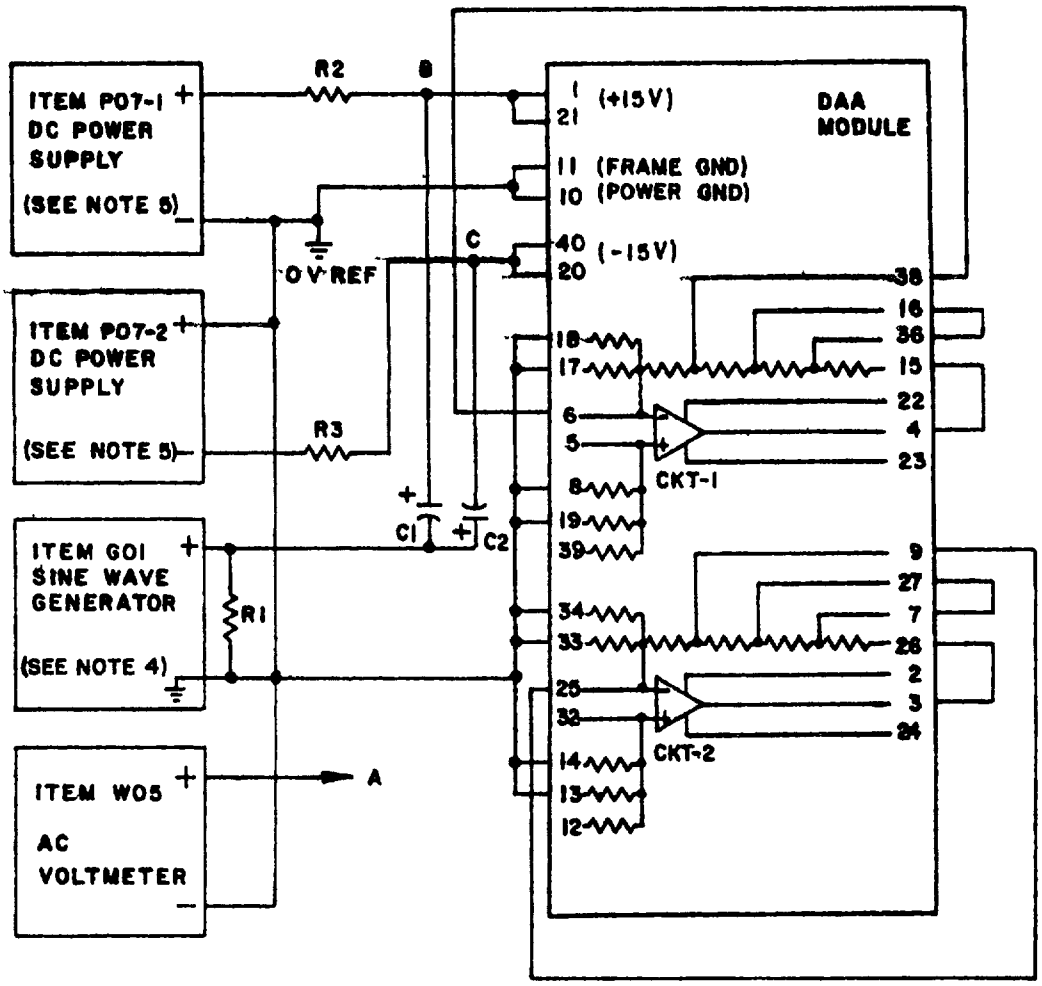
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TABLE XV. Power supply rejection ratio check test. 1/

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
T1	Power supply rejection ratio check	4	0	50	mV rms	19
T2	Power supply rejection ratio check	3	0	50	mV rms	19

1/ $PSRR = 20 \log \frac{(V_{OUT}/A_V)}{(V_{IN})} = 20 \log \left(\frac{\text{Item W05 reading}/20}{1.0 \text{ V rms}} \right)$.

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NOTES:
1. R1 = 49.9Ω ±5%, 1 W.
2. C1 = C2 = 20 µF ±20%, 20 Vdc electrolytic.
3. R2 = R3 = 100Ω ±5%, 1 W.
4. Item G01 setup:
   Frequency = 1 kHz ±5%.
   Amplitude 1.0 Vrms ±10% at points B and C.
5. Adjust P01-1 and P07-2 for ±15 V ±1% measured at points B and C respectively.

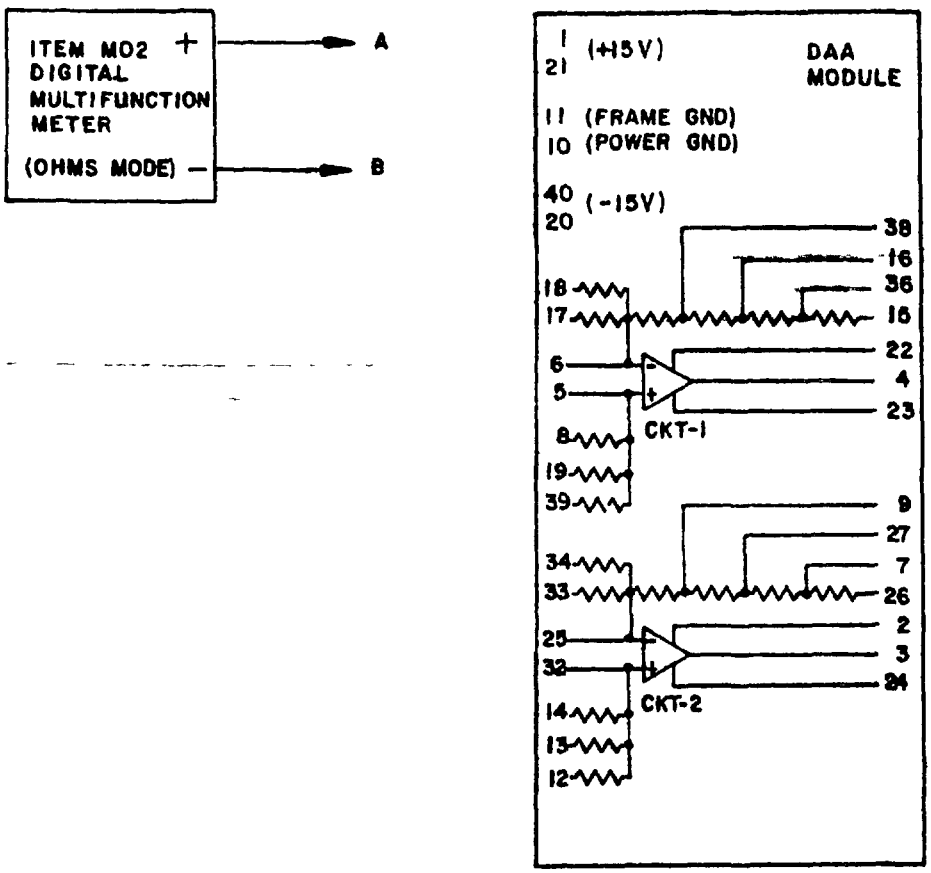
FIGURE 19. Power supply rejection ratio check test circuit for table XV.

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TABLE XVI. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	Continuity	11	Frame	0	1.0	Ω	20
U2	Continuity	1	21	0	1.0	Ω	20
U3	continuity	20	40	0	1.0	Ω	20

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NOTE: For test condition U1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 20 Continuity test circuit for table XVI.

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